

03 — Cardiovascular Flow Dynamics

BMED365 – Computational Medicine (Lab 5)

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1. Motivation & Learning Objectives

Why study cardiovascular flow dynamics?

The cardiovascular system is one of the most critical organ systems in the human body. The heart pumps approximately **5 liters of blood per minute** through a vast network of blood vessels spanning over **100,000 km** in total length. Understanding how blood flows through this network — how pressure waves propagate, how vessel compliance affects flow, and how pathological changes disrupt normal hemodynamics — is fundamental to modern medicine.

Cardiovascular disease (CVD) remains the leading cause of death worldwide, accounting for approximately 17.9 million deaths annually (WHO). Mathematical modeling of cardiovascular flow provides insight into disease mechanisms, guides surgical planning, and enables the design of medical devices such as artificial heart valves and stents.

From a computational perspective, cardiovascular flow modeling beautifully illustrates how the **same physical system** can be represented at multiple levels of abstraction — from simple lumped-parameter circuit analogies to full three-dimensional fluid–structure interaction simulations — each offering a different balance between simplicity, insight, and predictive power.

What you will learn

In this notebook you will:

1. **Understand** the biological basis of cardiovascular flow — how the heart generates pressure waves and how blood flows through the vascular tree
2. **Appreciate** the historical context — from William Harvey's discovery of blood circulation (1628) to modern computational fluid dynamics
3. **Master** the hierarchy of cardiovascular flow models — from simple lumped-parameter (Windkessel) models to advanced PDE-based models

4. **Implement** a lumped-parameter cardiovascular model in Python and simulate pressure and flow dynamics
5. **Visualize** and interpret pressure waveforms, flow rates, and their relationship to the ECG
6. **Explore** clinical implications — from hypertension to aortic stenosis to aneurysm mechanics

2. Historical Context: From Harvey to Computational Hemodynamics

The quest to understand blood flow

The understanding of blood circulation has a rich history spanning nearly two millennia, from Galen's ancient theories to modern patient-specific computational simulations:

Year	Milestone	Scientists
~170 AD	Galenic model: blood produced in liver, consumed by tissues	Galen of Pergamon
1242	First description of pulmonary circulation	Ibn al- Nafis
1628	<i>De Motu Cordis</i> — experimental proof of blood circulation	William Harvey
1733	First blood pressure measurement (in a horse!)	Stephen Hales
1828	Navier-Stokes equations for viscous fluid flow	Navier & Stokes
1840	Poiseuille's law for flow in cylindrical tubes	Jean Léonard Marie Poiseuille
1878	Pulse wave velocity measurements in arteries	Étienne-Jules Marey
1899	Windkessel (air-chamber) model of arterial compliance	Otto Frank
1955	Theory of pulsatile flow in elastic tubes	John R. Womersley
1960s	First computational models of arterial hemodynamics	several groups
1990s	Patient-specific CFD from medical imaging (CT/MRI)	C.A. Taylor and others
2000s–	Coupled multiscale models; 4D Flow MRI validation	multiple groups

William Harvey and the proof of circulation

Before Harvey, the prevailing view (from **Galen**, ~170 AD) was that blood was continuously produced in the liver, distributed through veins, consumed by tissues, and that arteries carried "vital spirits" mixed with air. This model had dominated Western medicine for nearly **1,500 years**.

William Harvey (1578–1657) demonstrated through careful experiments and quantitative reasoning that:

1. The heart acts as a **pump**, propelling blood through arteries
2. Blood circulates in a **closed loop** — arterial blood returns to the heart through veins
3. The **volume argument**: Harvey calculated that the heart pumps far more blood per hour than the body's total blood volume, proving the same blood must recirculate

4. **Valves** in veins ensure one-way flow back to the heart

"I began to think whether there might not be a motion, as it were, in a circle." — William Harvey, Exercitatio Anatomica de Motu Cordis et Sanguinis in Animalibus (1628)

Harvey's work is one of the great achievements in the history of physiology and set the stage for all subsequent understanding of cardiovascular function.

Otto Frank and the Windkessel model (1899)

The German physiologist **Otto Frank** (1865–1944) developed one of the first mathematical models of arterial blood flow. He observed that large arteries are **elastic** — they expand during systole (when the heart ejects blood) and recoil during diastole (when the heart relaxes). This behavior is analogous to the **Windkessel** (German for "air chamber") used in fire engines of the era — a closed air chamber that smooths pulsatile pump output into a more continuous stream.

Frank's insight that the **elasticity of large arteries** plays a crucial role in hemodynamics laid the foundation for all subsequent cardiovascular flow modeling. The Windkessel model remains widely used today as a simple but effective representation of arterial hemodynamics.



William Harvey (1578-1657)

William Harvey (1578-1657). Proved that blood circulates in a closed loop, with the heart as a pump.

Source: Wikimedia Commons (public domain).



2-Element Windkessel model

Otto Frank (1865-1944) developed the **Windkessel** (air-chamber) model of arterial compliance (1899), shown here as an electrical analog.

Diagram: Wikimedia Commons (CC BY-SA 4.0, K. Rachid).

3. The Cardiovascular System: Biological Background

The heart as a pump

The heart is a four-chambered muscular pump that generates the pressure needed to drive blood through the circulatory system:

- **Right atrium** → receives deoxygenated blood from the body via the vena cava
- **Right ventricle** → pumps blood to the lungs via the pulmonary artery
- **Left atrium** → receives oxygenated blood from the lungs via the pulmonary veins
- **Left ventricle** → pumps oxygenated blood to the entire body via the aorta

The left ventricle generates the highest pressures (~120/80 mmHg systolic/diastolic in a healthy adult) because it must push blood through the entire **systemic circulation**.

The vascular tree

The vascular system can be conceptualized as a branching tree with dramatically varying properties:

Vessel type	Diameter	Function	Typical pressure
Aorta	~25 mm	Windkessel effect; stores elastic energy	120/80 mmHg
Large arteries	4–25 mm	Conduct blood to organs; pulse wave propagation	110/75 mmHg
Arterioles	0.01–4 mm	Major site of resistance regulation	40–80 mmHg
Capillaries	5–10 μ m	Gas and nutrient exchange	15–35 mmHg
Venules / Veins	0.01–30 mm	Return blood to heart; capacitance vessels	5–15 mmHg
Vena cava	~30 mm	Final return path to right atrium	2–5 mmHg

The cardiac cycle

Each heartbeat (~0.8 s at 75 bpm) consists of two main phases:

1. **Systole** (~0.3 s): The ventricles **contract**, ejecting blood into the arteries

- Pressure in the left ventricle rises from ~5 to ~120 mmHg
- The aortic valve opens when LV pressure exceeds aortic pressure
- Blood is ejected at peak velocities of ~1 m/s into the aorta

2. **Diastole** (~0.5 s): The ventricles **relax** and fill with blood

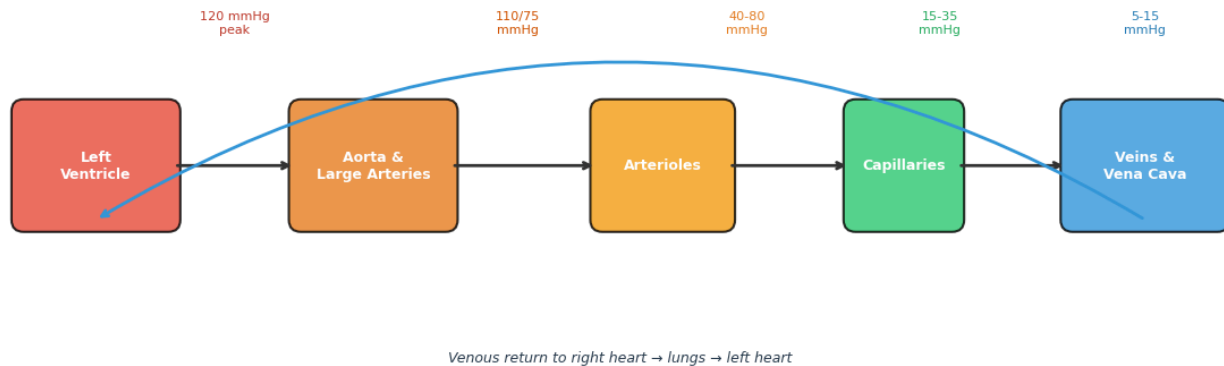
- Aortic valve closes → creates the **dicrotic notch** in the aortic pressure waveform
- Elastic recoil of arterial walls maintains pressure during diastole
- The heart fills passively, then via atrial contraction (the "atrial kick")

Key hemodynamic quantities

- **Cardiac output** (CO): Volume pumped per minute. $CO = \text{Heart Rate} \times \text{Stroke Volume} \approx 75 \text{ bpm} \times 70 \text{ ml} \approx 5.25 \text{ L/min}$

- **Blood pressure (BP)**: Force per unit area on vessel walls (measured in mmHg). Normal: 120/80 mmHg
- **Peripheral resistance (TPR)**: Opposition to flow, mainly in arterioles. Mean BP $\approx$ CO $\times$ TPR
- **Compliance (C)**: Ability of vessels to expand: $C = \Delta V / \Delta P$ (ml/mmHg)
- **Pulse wave velocity (PWV)**: Speed of pressure waves along arteries (4–10 m/s in humans; increases with arterial stiffness)

Schematic of the Systemic Circulation with Typical Pressures



4. Scientific and Clinical Implications

Understanding cardiovascular flow dynamics through mathematical modeling has had far-reaching consequences in both basic science and clinical medicine:

Cardiovascular Disease & Diagnostics

- **Hypertension** affects ~1.3 billion people worldwide. Lumped-parameter models help explain how increased peripheral resistance and reduced arterial compliance lead to elevated blood pressure
- **Atherosclerosis** — plaque buildup in arteries is associated with regions of **low and oscillatory wall shear stress**, which can be predicted by computational flow models
- **Aortic aneurysms** — weakened vessel walls lead to abnormal dilation; CFD simulations predict rupture risk based on wall stress distributions
- **Pulse wave velocity (PWV)** is a clinically validated biomarker of arterial stiffness and cardiovascular risk — directly related to vessel compliance in Windkessel models

Surgical Planning & Medical Devices

- **Coronary artery bypass grafting (CABG)**: Flow models help plan optimal graft placement and predict post-surgical hemodynamics
- **Stent design**: CFD simulations predict how stents affect local flow patterns and wall shear stress, informing device optimization

- **Artificial heart valves:** Prosthetic valve design relies heavily on flow modeling to minimize thrombosis and hemolysis
- **Ventricular assist devices (VADs):** Hemodynamic models guide device sizing, control algorithms, and patient selection

Advanced Diagnostics & Imaging

- **Fractional flow reserve (FFR):** Pressure measurements combined with flow models assess the functional significance of coronary artery stenosis — increasingly done *non-invasively* via CT-derived FFR (FFR_CT)
- **4D Flow MRI:** Combines velocity-encoded MRI with computational models for comprehensive, patient-specific hemodynamic assessment
- **Echocardiography** + flow models → non-invasive estimation of cardiac function, valve severity, and pulmonary pressures

Drug Development & Pharmacology

- Understanding how blood flow distributes drugs through the body (pharmacokinetics)
- Modeling how vasoactive drugs (e.g., nitroprusside, norepinephrine) alter hemodynamics by changing resistance and compliance parameters

Mathematical modeling of cardiovascular flow has evolved from simple analogies to patient-specific computational simulations that directly inform clinical decision-making.

Check Your Understanding: Cardiovascular Basics

Q1. Why does blood pressure *decrease* as it flows from large arteries to capillaries to veins?

► Click to reveal answer

Q2. What is the role of arterial **compliance** (elasticity) in the cardiovascular system? What would happen if arteries were rigid tubes?

► Click to reveal answer

Q3. If cardiac output is 5 L/min and mean arterial pressure is 93 mmHg (with venous pressure ≈ 0), what is the total peripheral resistance?

► Click to reveal answer

5. Modeling Cardiovascular Flow: From Windkessel to Navier-Stokes

The hierarchy of flow models

Cardiovascular flow can be modeled at different levels of complexity. Each level offers a different trade-off between **computational cost** and **physical fidelity**:

Level 0: Lumped-Parameter (0D) Models — The Windkessel Family

- Treat the vasculature as **discrete compartments** connected by resistances
- Use the **electrical circuit analogy** (pressure $\leftrightarrow$ voltage, flow $\leftrightarrow$ current, compliance $\leftrightarrow$ capacitance)
- Capture **global hemodynamics**: pressure, flow rate, and volume as functions of time
- **Cannot** resolve spatial flow patterns within vessels
- Computationally **very fast** — system of ODEs
- *Example*: The 2-, 3-, and 4-element Windkessel models

Level 1: One-Dimensional (1D) Models

- Resolve flow along the **length** of vessels (one spatial dimension)
- Use **cross-sectionally averaged** equations (area, flow rate, pressure as functions of position and time)
- Capture **pulse wave propagation**, wave reflections at bifurcations, and pressure/flow distribution in arterial networks
- Moderate computational cost — system of **hyperbolic PDEs**
- *Example*: 1D arterial network models (e.g., Olufsen et al., Sherwin et al.)

Level 2: Three-Dimensional (3D) CFD Models

- Solve the full **Navier-Stokes equations** in realistic 3D vessel geometries
- Resolve detailed **velocity fields**, wall shear stress, recirculation zones
- Can use **patient-specific geometry** from medical imaging (CT angiography, MRI)
- **High computational cost** — often requires HPC resources
- Essential for studying local flow phenomena near stenoses, bifurcations, and valves
- *Example*: Patient-specific aortic flow simulations, virtual FFR

Multiscale coupling

Modern approaches often **couple** models at different levels. For example, a 3D model of the aorta may be coupled with 1D models of downstream arteries and 0D (Windkessel) models as outlet boundary conditions. This provides both local detail where needed and global context.

The Windkessel models in detail

The Windkessel family of models progresses in complexity:

Model	Elements	Captures	Misses
2-element (RC)	Resistance R , Compliance C	Diastolic pressure decay	High-frequency impedance
3-element (RCR)	Adds characteristic impedance Z_0	Aortic input impedance at high frequencies	Inertial effects
4-element (RLCR)	Adds inertance L	Blood inertia; better high-frequency behavior	Spatial wave propagation

Two-Element Windkessel (Frank, 1899)

The simplest model — a single peripheral resistance R and arterial compliance C :

$$C \frac{dP}{dt} = Q_{\text{in}}(t) - \frac{P - P_{\text{venous}}}{R}$$

Three-Element Windkessel

Adds characteristic impedance Z_0 (representing the impedance of the proximal aorta) in series with the RC element:

$$P_{\text{aorta}} = Z_0 Q_{\text{in}} + P_C \quad ; \quad C \frac{dP_C}{dt} = Q_{\text{in}} - \frac{P_C}{R}$$

The Navier-Stokes equations for blood flow

At the most fundamental level, blood flow is governed by **conservation of momentum** (Navier-Stokes) and **conservation of mass** (continuity):

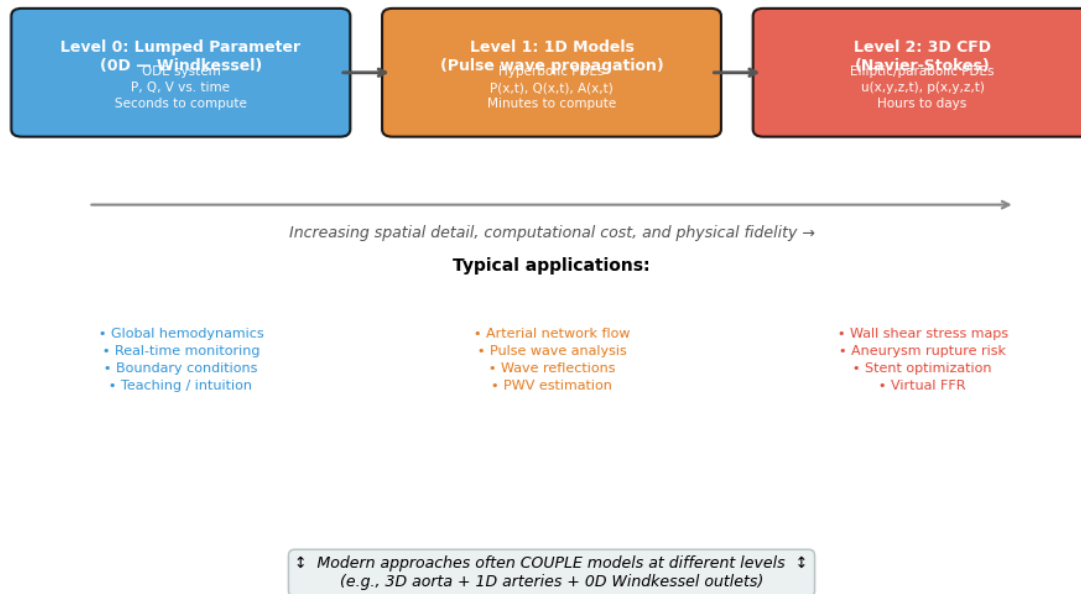
$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}$$

$$\nabla \cdot \mathbf{u} = 0$$

where $\rho \approx 1060 \text{ kg/m}^3$ (blood density), $\mathbf{u}$ is the velocity field, p is pressure, $\mu \approx 3.5 \times 10^{-3} \text{ Pa}\cdot\text{s}$ (dynamic viscosity), and $\mathbf{f}$ represents body forces.

For blood flow in elastic vessels, these equations must be coupled with models of **wall mechanics** — a challenging fluid–structure interaction (FSI) problem.

Hierarchy of Cardiovascular Flow Models



6. The Lumped-Parameter (Windkessel) Model: Conceptual Framework

The vessel wall as an electrical circuit

Just as Hodgkin and Huxley modeled the neural membrane as an electrical circuit (see Notebook 01), cardiovascular physiologists have long modeled the vascular system using an **electrical circuit analogy**:

Cardiovascular quantity	Electrical analog	Symbol	Unit
Pressure (P)	Voltage (V)	P	mmHg
Flow rate (Q)	Current (I)	Q	ml/s
Vascular resistance	Resistance (R)	R	mmHg·s/ml
Vessel compliance	Capacitance (C)	C	ml/mmHg
Blood inertance	Inductance (L)	L	mmHg·s ² /ml
Blood volume	Charge (q)	V	ml

This analogy is powerful because it allows us to use well-understood circuit theory to analyze hemodynamics!

Our model: Multi-compartment lumped-parameter network

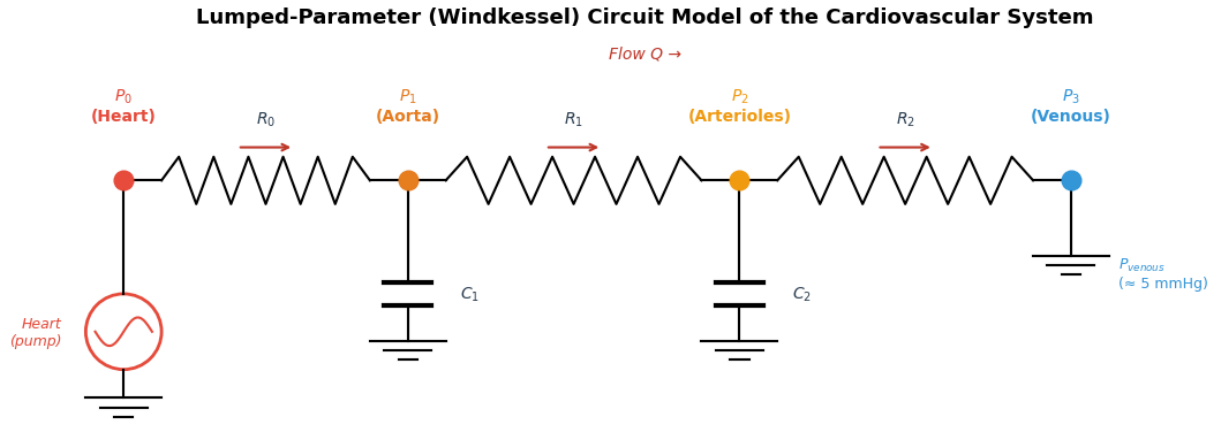
In this notebook, we model the cardiovascular system as a **series of connected compartments**, each representing a major section of the vasculature:

1. **Heart (P_0)**: A time-varying pressure source (the left ventricle)
2. **Compartment 1 (P_1)**: Represents the aorta and large arteries — with compliance C_1
3. **Compartment 2 (P_2)**: Represents smaller arteries and arterioles — with compliance C_2
4. **Venous system (P_3)**: Constant venous pressure (the "ground" of our circuit)

Adjacent compartments are connected by **resistances** (R_0, R_1, R_2) representing the viscous resistance to flow.

The electrical circuit analogy diagram

The heart acts as a **time-varying voltage source** (pressure generator), and each vascular compartment is a node with a capacitor (compliance) to ground. Resistors connect the nodes. See the programmatic diagram below for a clear visualization of this circuit.



7. Mathematical Formulation

Governing equations

For each compartment i with compliance C_i , the pressure dynamics are governed by conservation of mass (volume):

$$C_i \frac{dP_i}{dt} = Q_{\text{in},i} - Q_{\text{out},i}$$

where $Q_{\text{in},i}$ and $Q_{\text{out},i}$ are the inflow and outflow rates. Flow between adjacent compartments follows the **Ohm's law analog**:

$$Q_{i \rightarrow j} = \frac{P_i - P_j}{R_{i,j}}$$

System of equations for our model

Heart (pressure source):

$$P_0(t) = P_{\text{baseline}} + P_{\text{mean}} + A \sin\left(\frac{2\pi t}{T}\right)$$

where T is the cardiac cycle period, A is the pressure amplitude, and P_{baseline} sets the minimum diastolic pressure.

Compartment 1 (Aorta / large arteries):

$$C_1 \frac{dP_1}{dt} = \underbrace{\frac{P_0(t - \tau) - P_1}{R_0}}_{\text{inflow from heart}} - \underbrace{\frac{P_1 - P_2}{R_1}}_{\text{outflow to arterioles}}$$

Compartment 2 (Arterioles / small vessels):

$$C_2 \frac{dP_2}{dt} = \underbrace{\frac{P_1 - P_2}{R_1}}_{\text{inflow from aorta}} - \underbrace{\frac{P_2 - P_3}{R_2}}_{\text{outflow to veins}}$$

where τ is a propagation delay representing the finite speed of pressure wave travel, and P_3 is a constant venous pressure.

ECG signal model

The ECG is modeled as a sum of Gaussian functions representing the cardiac electrical events:

$$\text{ECG}(t) = \text{baseline} + P_{\text{wave}}(t) + \text{QRS}(t) + T_{\text{wave}}(t)$$

Each component is a Gaussian (or sum of Gaussians):

$$\text{component}(t) = A_k \exp\left(-\frac{(t_{\text{cycle}} - t_k)^2}{2\sigma_k^2}\right)$$

The ECG and pressure signals are coupled through their timing relationships: the QRS complex (ventricular depolarization) precedes the ventricular pressure rise by a few tens of milliseconds.

Model parameters

Parameter	Value	Physiological meaning
R_0	0.1 mmHg·s/ml	Aortic valve resistance (low)

Parameter	Value	Physiological meaning
R_1	0.8 mmHg·s/ml	Large artery resistance
R_2	2.0 mmHg·s/ml	Peripheral (arteriolar) resistance (high)
C_1	1.5 ml/mmHg	Aortic/arterial compliance
C_2	2.0 ml/mmHg	Arteriolar compliance
T	1.0 s	Cardiac period (60 bpm)
τ	0.1 s	Pressure wave propagation delay
P_3	5.0 mmHg	Venous pressure (constant)

Numerical solution

The system is integrated using forward Euler:

$$P_i(t + \Delta t) = P_i(t) + \Delta t \cdot \frac{dP_i}{dt}$$

with time step $\Delta t = 0.01$ s (sufficient for the smooth pressure waveforms at 60 bpm).

Check Your Understanding: The Mathematics

Q4. In the lumped-parameter model, what physical property does the **capacitor** (compliance C) represent? What happens to the pressure waveform in compartment 1 if you *double* C_1 ?

► Click to reveal answer

Q5. Why is the resistance R_2 (peripheral resistance) the **largest** in our model? What is its physiological basis?

► Click to reveal answer

Q6. What is the role of the propagation delay $\tau = 0.1$ s in the model? Is this physiologically realistic?

► Click to reveal answer

8. Python Implementation: Lumped-Parameter Cardiovascular Model

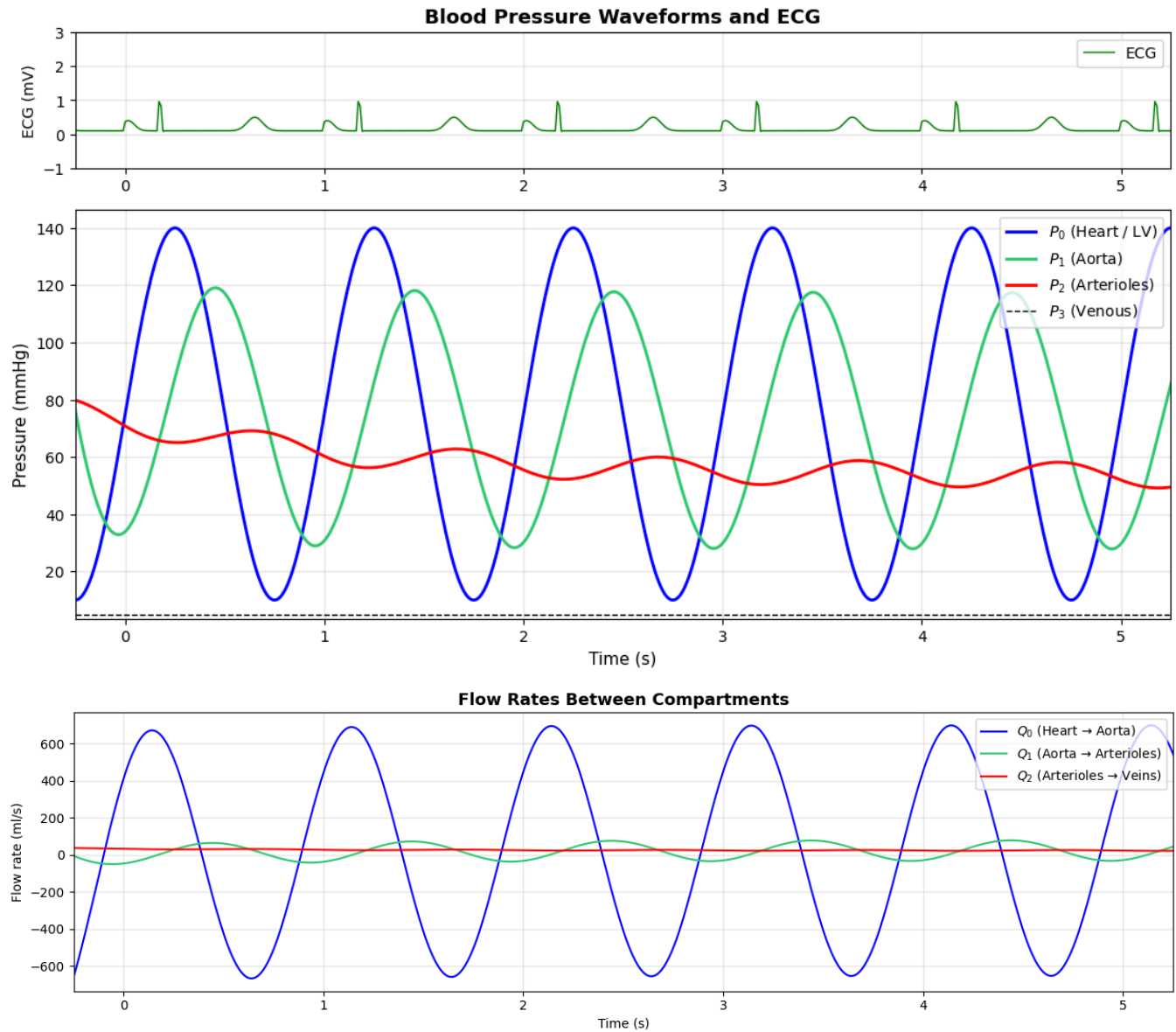
We implement the model as a Python class `CardiovascularModel` that:

1. Defines the heart as a time-varying pressure source

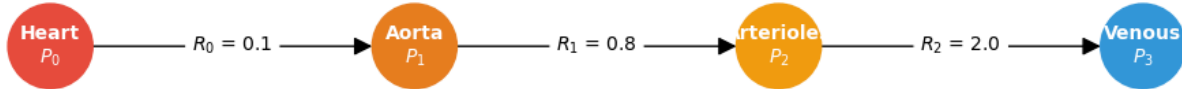
2. Generates a synthetic ECG signal synchronized with the pressure
3. Solves the lumped-parameter ODE system using forward Euler integration
4. Stores time series of all pressures and the ECG for visualization

/var/folders/1g/_4x26lps6megg1tr73d7g2bmh0000gp/T/ipykernel_25266/3248090389.py:35: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.

```
plt.tight_layout()
```



Network Model: Lumped-Parameter Cardiovascular Circuit



Interpreting the simulation results

ECG signal (top panel):

- Shows normal cardiac electrical activity with clear **P waves** (atrial depolarization), **QRS complexes** (ventricular depolarization), and **T waves** (ventricular repolarization)
- Heart rate: 60 bpm (1 beat per second)
- The QRS complex precedes the ventricular pressure rise — reflecting the normal **electro-mechanical coupling delay**

Pressure waveforms (bottom panel):

Signal	Represents	Key features
P₀ (blue)	Left ventricular pressure	Oscillates 5–140 mmHg; sharp systolic rise and diastolic fall
P₁ (green)	Aortic / large artery pressure	Smoothed and delayed version of P ₀ ; compliance C ₁ damps the pulsations
P₂ (red)	Arteriolar pressure	Much less pulsatile; mean ~60 mmHg; further damped by C ₂ and R ₂
P₃ (dashed)	Venous pressure	Constant at 5 mmHg (our simplifying assumption)

Flow rates:

- Q_0 (Heart → Aorta) shows the largest pulsatility — driven directly by the heart
- Q_1 and Q_2 become progressively smoother as compliance and resistance damp the pulsations
- Notice how compliance acts as a **low-pass filter**, converting pulsatile cardiac output into smoother downstream flow

Network graph: Shows the four-compartment lumped-parameter model with resistance values connecting the nodes.

Check Your Understanding: Simulation Results

Q7. Why does the P_1 (aortic) pressure waveform look "smoother" (less pulsatile) than the P_0 (ventricular) pressure waveform?

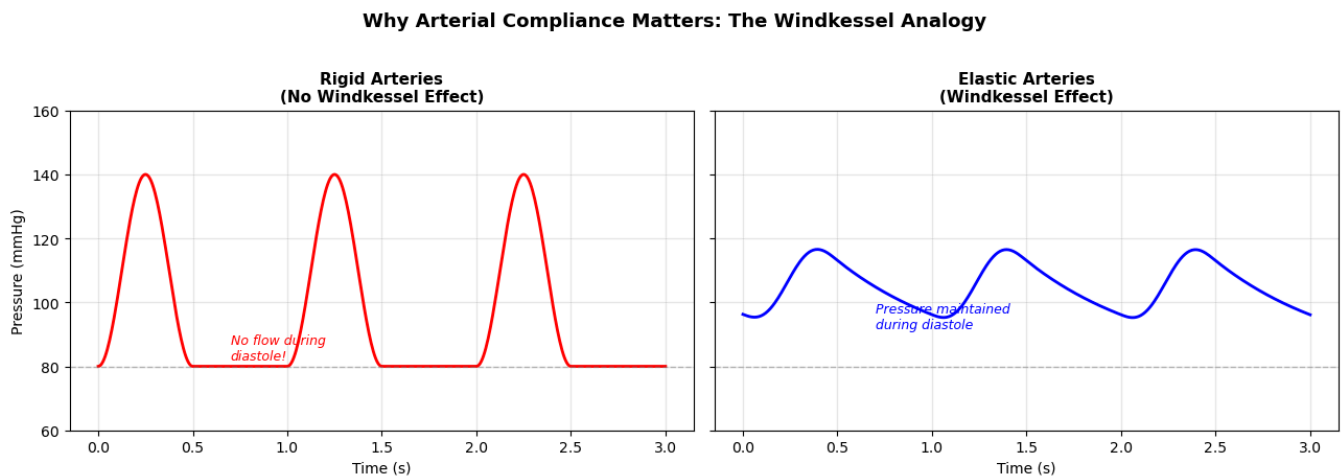
► Click to reveal answer

Q8. The model uses forward Euler integration with $\Delta t = 0.01$ s. What could go wrong if you use a much larger time step, say $\Delta t = 0.5$ s?

► Click to reveal answer

Q9. In the pressure plot, there is an initial "transient" phase where the pressures are settling. Why does this happen, and roughly how long does it take to reach steady-state oscillation?

► Click to reveal answer



DIY Exercises: Explore the Lumped-Parameter Model

Exercise 1: Simulate Hypertension — Increased Peripheral Resistance

In hypertension, arteriolar smooth muscle contraction increases peripheral resistance. Simulate this by increasing R_2 from 2.0 to 4.0 mmHg·s/ml.

► Click for starter code

► Click to reveal what you should see

Exercise 2: Simulate Aortic Stiffening (Aging / Arteriosclerosis)

With aging, large arteries lose elasticity (compliance decreases). Simulate this by reducing C_1 from 1.5 to 0.3 ml/mmHg.

► Click for starter code

► Click to reveal what you should see

Exercise 3: Explore the Effect of Heart Rate

Change the cardiac period T from 1.0 s (60 bpm) to 0.5 s (120 bpm, as during exercise). What happens to the pressure waveforms?

- Click for starter code
- Click to reveal what you should see

9. Beyond Lumped Parameters: Toward 1D and 3D Models

1D Pulse Wave Propagation

The lumped-parameter model cannot capture **spatial** aspects of blood flow — where in the aorta is the pressure highest? How do pulse waves reflect at bifurcations? For this, we need at least a **1D model**.

The 1D equations for flow in an elastic tube (cross-sectionally averaged) are:

$$\frac{\partial A}{\partial t} + \frac{\partial(Au)}{\partial x} = 0 \quad (\text{conservation of mass})$$
$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = -\frac{f}{\rho A} \quad (\text{conservation of momentum})$$

where $A(x, t)$ is the cross-sectional area, $u(x, t)$ is the velocity, $p(x, t)$ is the pressure, and f represents viscous friction. These are closed by a **tube law** relating pressure to area:

$$p - p_{\text{ext}} = \frac{\beta}{A_0} \left(\sqrt{A} - \sqrt{A_0} \right), \quad \beta = \frac{\sqrt{\pi} h E}{1 - \nu^2}$$

where E is the Young's modulus of the vessel wall, h is wall thickness, A_0 is the reference area, and ν is the Poisson ratio.

These equations form a **hyperbolic system** that supports wave propagation at the **pulse wave velocity**:

$$c = \sqrt{\frac{\beta}{2\rho A_0}} \cdot A^{1/4}$$

The code cell below demonstrates a simplified 1D pulse wave propagation in a single elastic tube.

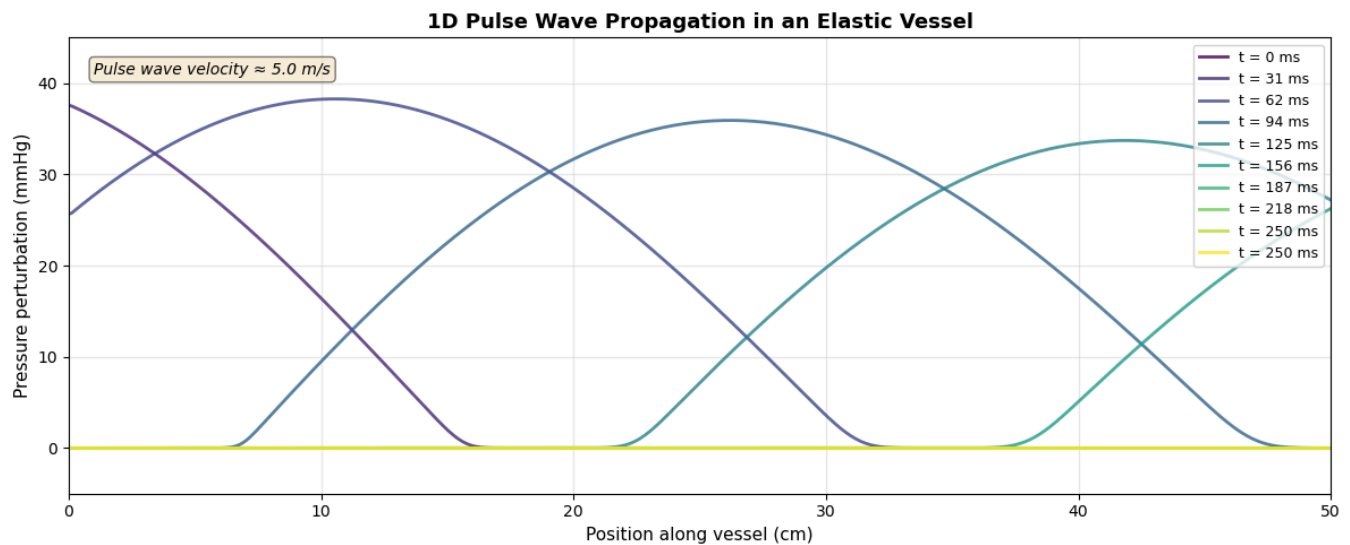
3D Computational Fluid Dynamics (CFD)

For detailed local flow fields — especially near stenoses, bifurcations, and valves — the full **3D Navier-Stokes equations** are solved, often with:

- **Patient-specific geometry** reconstructed from CT or MRI scans
- **Fluid-structure interaction** (FSI) to model wall deformation

- **Non-Newtonian** blood rheology for accurate shear-dependent viscosity
- **Turbulence models** for high-Reynolds-number flows (e.g., post-stenotic jets)

These simulations typically require high-performance computing and specialized software (e.g., SimVascular, CRIMSON, OpenFOAM, COMSOL).



10. Going Deeper

Extensions of the lumped-parameter model

The basic model presented here can be extended in many directions:

- **More compartments:** Add pulmonary circulation, coronary circulation, cerebral circulation
- **Heart valves:** Model valves as diodes (one-way flow) with opening/closing dynamics
- **Inertance:** Add inductors (L) to capture blood inertia effects → 4-element Windkessel
- **Baroreceptor reflex:** Implement a feedback controller that adjusts heart rate and peripheral resistance based on arterial pressure
- **Ventricular pressure-volume loops:** Replace the sinusoidal P_0 with a time-varying elastance model for more realistic ventricular mechanics
- **Pathological conditions:** Simulate aortic regurgitation, mitral stenosis, ventricular septal defect

Key references

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2. **Quarteroni, A., Manzoni, A., & Vergara, C.** (2017). The cardiovascular system: Mathematical modelling, numerical algorithms and clinical applications. *Acta Numerica*, 26, 365–590. [Comprehensive review of all modeling levels]
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1D approaches]

4. **Taylor, C.A. & Figueroa, C.A.** (2009). Patient-specific modeling of cardiovascular mechanics. *Annual Review of Biomedical Engineering*, 11, 109–134. [Patient-specific CFD]
5. **Nichols, W.W., O'Rourke, M.F., & Vlachopoulos, C.** (2011). *McDonald's Blood Flow in Arteries*. 6th edition. [The classic textbook on arterial hemodynamics]

Useful resources

- **SimVascular** — Open-source cardiovascular simulation platform: <https://simvascular.github.io/>
- **OpenBF** — Open-source 1D blood flow solver in Julia: <https://github.com/INSIGNEO/openBF>
- **Pulse wave hemodynamics** (review article, open access):
<https://doi.org/10.1152/ajpheart.00218.2021>
- **Cardiovascular Mathematics** (Formaggia, Quarteroni, Veneziani) — comprehensive textbook on mathematical modeling of the cardiovascular system